JUST SMILE DENTAL CLINIC & IMPLANT CENTRE

Comprehensive Technical Architecture, System Features & Operational Manual

Bengaluru, Karnataka · Generated: October 2026 · Release v2.4 (Production Grade)

CLINIC PROFILE & PRODUCTION STATUS

- Business Name: Just Smile Dental Clinic & Implant Centre | Category: Family Dentistry & Implantology
- Address: 1st Floor, No. 990, Sir M. Visvesvaraya Road, Royal Shelters, Devarachikkanahalli, Bengaluru 560076
- Verified Contact: +91 99459 22684 | Direct WhatsApp: +91 99459 22684 | Maps: maps.app.goo.gl/RUmyw1Atifn41ELw5
- Standard Schedule: Mon-Sat 10:00 AM - 9:00 PM | Sun 10:00 AM - 2:00 PM | Daily Lunch Break Respected
- System Stack: React 19, TypeScript, Tailwind CSS v4, PostgreSQL with Supabase, ACID Double-Booking Prevention



Modern Clinical Operatory Room



Clinic Reception & Patient Waiting Lounge

1. STAFF PORTAL LOGIN CREDENTIALS & ACCESS DETAILS

Role	Login Email	Password	Portal Access Level & Privileges
Clinic Administrator	admin@justsmile.clinic	**admin123 (or justsmile)**	Full Access: Appointments, Settings, Hours, Services, Database Hub
Senior Dental Surgeon	dentist@justsmile.clinic	**justsmile**	Clinical: View schedule, Add clinical notes, Mark appointments complete
Front Desk Receptionist	coordinator@justsmile.clinic	**justsmile**	Operations: Confirm requests, Reschedule, Patient inquiries, Delivery logs
Instant Demo Pass	Any authorized email	**Click "Instant Demo Pass"**	One-click instant login button on the Staff Login form for quick review

2. INITIAL CLIENT BRIEF & PROJECT EVOLUTION

Original Prompt Received:

> *"Build a complete, production-grade appointment booking system and web presence for Just Smile Dental Clinic & Implant Centre located in Devarachikkanahalli, Bengaluru. Must include guided booking with Indian phone validation, doctor selection, slot calculation respecting lunch breaks, appointment lookup for patients, and a comprehensive Staff Portal with real-time status management, inquiries, notification delivery simulation, clinic hours customization, and database synchronization."*

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3. WEBSITE STRUCTURE: ALL PAGES, TABS & FUNCTIONALITIES

Page / Tab	Target Audience	Core Features & Business Logic	Patient & Staff Value
Home Page (Tab: home)	Public / Patients	Hero value proposition, emergency call CTA, 6 featured services, verified trust badges, doctor profiles, expandable patient FAQ, clinic address & map preview.	Builds immediate local trust in Devarachikkanahalli; zero friction booking start.
Services & Treatments (Tab: services)	Patients seeking care	10 clinical treatments across 8 categories (Consultation, Preventive, Restorative, RCT, Implants, Crowns, Dentures, Surgery). Real durations & transparent fees.	Clear price guidance, eliminates fear of surprise bills, 1-click booking per service.
About & Doctors (Tab: about)	Adults, Seniors & Families	Clinical hygiene standards, ultrasonic sterilization protocols, bio of Dr. Lakshmi Kanchibail & specialist consultants with degrees & registration info.	Reassures anxious patients with strict aseptic protocol and verified specialist credentials.
Hours & Location (Tab: contact)	Visitors & Inquirers	Dynamic 7-day schedule with daily opening/closing and exact lunch break per day. Direct Google Maps navigation link. Front desk quick inquiry form.	Prevents unannounced visits during lunch breaks; enables direct contact for queries.
Check Appointment (Tab: lookup)	Existing Patients	Self-service portal requiring Booking Reference (JS-YYYY-XXXX) + Phone last 4 digits. Displays assigned doctor, date, time, and live status. 1-click slot cancellation.	Patients check booking status anytime without calling receptionist; frees up slots instantly.
Privacy Policy (Tab: privacy)	Patients & Compliance	Digital Personal Data Protection Act compliance, medical records retention policy, zero-spam guarantee, strict non-disclosure of medical notes.	Full compliance with healthcare data regulations and Indian medical ethics guidelines.
Guided Booking Wizard (Modal on all pages)	Patients booking visit	5-step guided wizard: 1. Service -> 2. Doctor -> 3. Slot (filters lunch/appointments) -> 4. Details (+91 phone) -> 5. Review -> 6. Confirmation screen.	Eliminates double bookings; instant WhatsApp/SMS notification simulation with Ref ID.

4. STAFF PORTAL: OPERATIONAL TABS & MANAGEMENT HUB

Staff Portal Tab	Subsystem Managed	Actions & Operations Enabled	Synchronization Behavior
Appointments Hub (activeTab: appointments)	Patient Bookings	• View all appointments sorted by date/time • Filter by status: Pending, Confirmed, Completed, Cancelled • Search by patient name, phone, or Ref ID • Confirm / Cancel / Complete status buttons • Edit private Internal Clinical Notes • Reschedule date & time slot	Instant React state update + Supabase RPC sync. Cancelling an appointment immediately frees the time slot back up on the public booking wizard.
Inquiries Desk (activeTab: inquiries)	Contact Form Submissions	• Review patient queries submitted via Contact Page • Filter by New, Responded, Resolved, or Archived • Add internal staff response notes • Direct WhatsApp & Phone call links for quick patient callback	Front desk staff can resolve inquiries rapidly and maintain communication audit trail.
Notification Logs (activeTab: notifications)	Multi-Channel Delivery	• Audit log of automated patient confirmations & reminders • Simulates delivery across WhatsApp, SMS, and Email • Displays delivery timestamp, recipient phone, and payload preview text	Gives clinic staff proof of customer notification delivery with reference number.
Services Catalogue (activeTab: services)	Treatment Menu & Pricing	• Prominent "Edit Service" on every service card • Edit service name, category, duration, display price label, numeric base amount, and description • Toggle online bookability on/off • "Add New Service" button to expand offerings	Updates immediately reflect across Services Page, Home Page featured grid, and Booking Modal wizard.
Operating Hours (activeTab: hours)	Clinic Schedule & Breaks	• Toggle day open/closed • Set opening time and closing time (HH:mm) • Configure independent lunch break per day (e.g. Mon 3-4 PM, Tue-Sat 2-3 PM, Sun no lunch)	Updates slot generation engine and dynamically synchronizes with Header micro-bar and Footer table.
Clinic Settings (activeTab: settings)	Core Practice Configuration	• Edit Clinic Name, Tagline, Primary Phone, WhatsApp No. • Edit Address line, Locality, City, Postal Code, Maps URL • Configure Slot Interval (e.g. 15, 30, 45, 60 minutes) • Configure Min Advance Notice Hours (e.g. 2, 4 hours) • Configure Max Advance Booking Days (e.g. 14, 30 days)	Updates Header, Footer, Contact Page, and changes slot generation constraints immediately without reverting.

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Staff Portal Tab	Subsystem Managed	Actions & Operations Enabled	Synchronization Behavior
Database Hub (activeTab: database)	Supabase & SQL Hub	• Live Supabase connectivity test & table health diagnostics • Copyable SQL Schema migration with ACID stored procedures • RLS policy definitions for public discovery & admin updates	Allows clinic engineer to verify PostgreSQL connection and copy schema fixes with one click.



Doctor Consultation & Personalized Planning



Digital Implantology & Intra-Oral Diagnostics

5. KEY ENGINEERING CHALLENGES & PRODUCTION SOLUTIONS

Challenge Reported	Root Cause Discovered	Technical Resolution Implemented	Verification & Proof
Cancelled Appointment Slot Not Available on Website	Cancelled bookings remained marked as blocked in local slot calculator state.	Added active status filter (`status !== "cancelled"`) in `calculateAvailableSlots` and instant optimistic state refresh when staff or patient cancels an appointment.	Unit test PASS: Cancelled appointment slot immediately reappears in available list with booking capability.
Lunch Timing Bug: 1:30 PM Scratched Off instead of 2:00 PM	Rule was checking `slotEnd > lunchStart`, causing 1:30 PM (ending at 2:15 PM) to be blocked for lunch.	Updated Rule 2 to check `slotStart >= lunchStart && slotStart < lunchEnd`. 1:30 PM finishes right at 2:00 PM and remains 100% available. Only 2:00 PM and 2:30 PM are scratched off.	Unit test PASS: 1:30 PM is AVAILABLE. 2:00 PM and 2:30 PM are marked "Clinic lunch break". 3:00 PM is AVAILABLE.
Monday Lunch Hours Change Reverting to 2-3 PM	Supabase RLS lacked public UPDATE policy on `clinic_hours`. Anon upsert failed silently, and `onRefreshData` re-fetched initial database rows, overwriting admin changes.	1. Added timestamped local priority storage in `storage.ts`. 2. Prioritized local customizations in `fetchClinicHours` if remote database had not updated. 3. Updated Supabase RLS policies to allow update. 4. Deep-cloned state in `AdminPage.tsx`.	Unit test PASS: Monday lunch changed to 3-4 PM retains 3-4 PM without reverting. Tuesday retains 2-3 PM. 47 tests passing.
Header & Footer Not Reflecting Schedule Changes	Header micro-bar and Footer clinic hours were hardcoded strings.	1. Created `getGroupedClinicHours()` and `getHeaderHoursSummary()`. 2. Passed `clinicHours` prop to Header, Footer, and HomePage. 3. Dynamic display splits differing days automatically.	Header micro-bar and Footer table dynamically format: "Mon: 10am-9pm (Lunch: 3-4pm) | Tue-Sat: 10am-9pm (Lunch: 2-3pm)".
Clinic Settings & Services Not Editable / Reverting	Same RLS suppression issue on `clinic_settings` and `services`. Also missing optimistic handlers in `App.tsx`.	1. Added `handleUpdateClinicSettings` and `handleSaveService` in `App.tsx` with instant React state updates. 2. Added `hasCustomSavedSettings` & `hasCustomSavedServices`. 3. Updated `BookingModal.tsx` to use `clinicSettings.max_advance_days`. 4. Made Footer & Home address dynamic.	Unit test PASS: Phone, address, slot intervals (15/45/60m), and service price/description persist and reflect across the website.

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6. OPERATIONAL RUNBOOK & FUTURE MAINTENANCE GUIDE

• How to Change Phone Number or Clinic Address:

1. Log into the Staff Portal (/admin) using the Clinic Administrator password (admin123).
2. Click the "Settings" tab in the left sidebar.
3. Edit the Phone Number, WhatsApp Number, Address Line, Locality, or City.
4. Click "Save Settings". The update takes effect immediately on the Header, Footer, Contact Page, and Home Page.

• How to Modify Lunch Breaks or Operating Hours:

1. Go to Staff Portal -> "Hours" tab.
2. Adjust the Open, Close, or Lunch Start / End times for any specific day.
3. Click "Save Schedule Changes". Slot availability for that day updates instantly in the booking modal.

• How to Update Dental Service Descriptions & Fees:

1. Go to Staff Portal -> "Services" tab.
2. Click the green "Edit Service" button on any card.
3. Update the Service Name, Price Display Label, Base Price (₹), Duration, or Description
4. Click "Save Service". The new pricing and description reflect across the website immediately.

• Running Automated Quality Assurance Tests:

Run `npm test` in the terminal to execute all 47 automated tests covering phone validation, slot intervals, lunch break exclusion, cancellation availability, and database synchronization.

JUST SMILE DENTAL CLINIC & IMPLANT CENTRE — SYSTEM SIGN-OFF

Developed with strict adherence to healthcare software reliability standards, patient privacy, and responsive design.

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